

BAYES' THEOREM - SIMPLE EXPLANATION

Bayes' theorem updates probability when new evidence appears.

BEFORE EVIDENCE

**Initial belief
called PRIOR**

NEW EVIDENCE

**Something new
is observed**

AFTER EVIDENCE

**Updated belief
called POSTERIOR**

Old probability + new evidence = updated probability

STEP 1: START WITH 10,000 TRANSACTIONS

Fraud is rare: only 1% of transactions are fraudulent.

FRAUDULENT

**100 transactions
1% of total**

GENUINE

**9,900 transactions
99% of total**

Before any alert: $P(\text{Fraud}) = 100 / 10,000 = 1\%$

STEP 2: THE SYSTEM GENERATES ALERTS

The system catches fraud, but it also creates false alerts.

FRAUD ALERTS

90 of 100 frauds
trigger an alert

FALSE ALERTS

495 of 9,900 genuine
transactions alert

TOTAL ALERTS

90 + 495
585 alerts

An alert is evidence - but it is not proof of fraud.

STEP 3: LOOK ONLY AT THE ALERTS

Among 585 alerts, how many are truly fraudulent?

$$\begin{aligned} P(\text{Fraud} \mid \text{Alert}) &= \text{Fraud alerts} / \text{All alerts} \\ &= 90 / 585 = 15.4\% \end{aligned}$$

After seeing the alert, fraud probability rises from 1% to 15.4%.

BAYES' THEOREM FORMULA

$$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$$

P(A)

Prior probability

P(B | A)

Likelihood of evidence

P(A | B)

Updated probability

REMEMBER THE MAIN IDEA

An alert changes what we believe, but it does not prove the event.

BEFORE ALERT

Fraud probability
1%

AFTER ALERT

Fraud probability
15.4%

Bayes asks: Given the evidence, what is now most likely?